

Coding & Solution

Date	15 July 2026
Team ID	*****
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented code and overall solution.

Solution Summary

Field	Details
Repository Link / URL	(Paste your GitHub repository link here after uploading the project.)
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Key Features Implemented	• Responsive Home Page for project introduction. • Loan Application Form for collecting applicant details. • Data preprocessing including missing value handling, label encoding, feature scaling, and SMOTE balancing. • Multiple Machine Learning models implemented (Decision Tree, Random Forest, KNN, Gradient Boosting). • Best-performing Random Forest model selected for prediction. • Loan approval result displayed on a separate result page using Flask.
Pending / Incomplete Features	• User login and authentication. • Database integration for storing applicant records. • Admin dashboard and application history. • Cloud deployment (AWS/IBM Cloud/Render) and REST API integration.
Setup / Run Instructions	1. Install required packages: pip install flask pandas numpy scikit-learn matplotlib seaborn imbalanced-learn

	2. Run the application: python app.py 3. Open the localhost URL displayed in the terminal. 4. Click "Start Prediction", enter the applicant details, and submit the form to view the loan approval prediction.
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Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Update after pushing the project to GitHub

Additional Notes / Comments

The **SmartLender** application was successfully implemented and tested from end to end. The workflow begins with the applicant entering loan details through the web interface. The backend preprocesses the input data by handling missing values, encoding categorical features, scaling numerical features, and preparing the feature vector. The trained **Random Forest** machine learning model predicts whether the loan application is likely to be approved or rejected. The prediction result is displayed instantly on the result page through the Flask web application. The system provides fast, consistent, and reliable loan approval predictions, demonstrating the practical application of machine learning in the banking and financial sector.